Empirical Proof Record: VALIDATED

2026-05-20T15:34:50.234075+00:00

Empirical Proof Record — VALIDATED

Proof ID: `0def893eb9321562a14d9541929b3a07…`
Created: 2026-05-20T15:34:50.234075+00:00

1. Claim

> Across 30 synthetic two-sample pairs of size 50 per group with varying effect sizes and variance ratios (seed=20260518), "scipy.stats.ttest_ind(equal_var=False)", "statsmodels.stats.weightstats.ttest_ind(usevar='unequal')", and "pingouin.ttest(correction=True)" produce Welch t-statistics AND two-sided p-values that agree to ≤ 1e-09 across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way variant with genuinely-independent statsmodels.)

- **Operationalisation:** Generate N (x, y) pairs where x ~ Normal(0, σ_x^2) and y ~ Normal(δ, σ_y^2) for a sweep of δ across [-1.0, 1.0] and variance-ratio σ_y/σ_x across [0.5, 2.0]. Compute Welch's t and two-sided p under each backend. Compute max_abs_diff = max over (pair, statistic ∈ {t, p}, backend-pair) of |stat_a - stat_b|. VALIDATED iff max_abs_diff ≤ tolerance.
- **Threshold:** `max_absolute_welch_difference <= 1e-09 t_or_p`
- **H0:** At least one backend pair disagrees on the Welch t-statistic OR the two-sided p-value by more than the floating-point tolerance.
- **H1:** All three backends compute identical Welch t AND p to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed `1.7763568394002505e-15`
30 pairs × 3 backends × 2 statistics; max pairwise |Δ| = 1.776e-15 on (t, scipy_vs_statsmodels) (≤ 1e-09 tol)

3. Evidence

Pillar	Statistic	
welch_t_scipy_vs_statsmodels_vs_pingouin	max_absolute_welch_difference	1.776356839400250

4. Pre-registration

- Registered at: `2026-05-20T15:34:50.003601+00:00`
- Config hash: `650f785bb6c4c650aa559236e9e29db3…`
- Data hash: `650f785bb6c4c650aa559236e9e29db3…`

5. Signature

`7eb7781079344eff11878e9669edf470b0e0866a50dfe7c3dc654664818b5ac5`